

Resampling Methods

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Topics to cover

- The bootstrap principle.
 - Nonparametric versus parametric bootstraps.
 - Bias reduction
 - Confidence intervals
 - Null distribution of test statistics

Example of bootstrap inconsistency

- Exact and asymptotic pivotal statistics
- The edgeworth expansion and higher order precision.
- Subsampling.

Reference books:

- Chapter 10, Bruce Hansen's econometrics book
- Hall, Peter, The bootstrap and Edgeworth expansion, 2013, Springer
<https://link.springer.com/book/10.1007/978-1-4612-4384-7>
- Politis, D and Romano, J and Wolf, M, Subsampling, 2012, Springer
<https://link.springer.com/book/10.1007/978-1-4612-1554-7>

- Now we formally look at the statistical bootstrap principle.
- Replace the real world by the bootstrap world:
 - Real World: Population(F_0) $\longrightarrow$ Sample(F_1): $X_1, \dots, X_n$.
 - The bootstrap world: Sample(F_1): $X_1, \dots, X_n \longrightarrow$ Bootstrap Sample $F_2 = X_1^*, \dots, X_n^*$.
- We care about functional of $F_0 : \theta(F_0)$, the bootstrap principle says that we estimate $\theta(F_0)$ by $\theta(F_1)$.
- The only problem is how to define $\theta(F_0)$, and the bootstrap resample is only useful for defining this function for $\theta(F_1)$.
- A bootstrap resample is a sample of size n , drawn independently with replacement from the empirical distribution F_1 , i.e.,

$$P(X_i^* = X_j | F_1) = n^{-1}, \quad 1 \leq i, j \leq n.$$

- The simplest example: the mean.

$$\theta(F_0) = \mu = \int x dF(x).$$

The bootstrap estimate is

$$\theta(F_1) = \int x dF_1(x) = \frac{1}{n} \sum_{i=1}^n X_i = E(X_i^* | F_1)$$

- Similarly, for the variance.

$$\begin{aligned} \theta(F_0) = \sigma^2 &= \int x^2 dF(x) - \left(\int x dF(x) \right)^2 \\ \theta(F_1) = \hat{\sigma}^2 &= \int x^2 d\hat{F}(x) - \left(\int x d\hat{F}(x) \right)^2 \\ &= E(X_i^{*2} | F_1) - (E(X_i^* | F_1))^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - (\bar{X})^2 \end{aligned}$$

- Drawing X_i^* from F_1 is called nonparametric bootstrap.
- In regression models, $y_i = x_i' \beta + \epsilon_i$, the nonparametric bootstrap (for estimating the distribution of $\hat{\beta}$, say) draws (y_i^*, x_i^*) from the JOINT empirical distribution of (y_i, x_i) . In statistics this is called the correlation model.
- In a regression model with fixed design of regressors, draw instead from $\hat{\epsilon}_i = y_i - x_i' \hat{\beta}$ fixing the x_i 's.
- In a correctly specified conditional mean regression model they are asymptotically equivalent.
- With d dimension data you can find many different ways of resampling, depending on your assumptions about the relation among y_i, x_i , for example.
- You can also modify your bootstrap resample scheme by taking into account a priori information you have about X_i , say if you know X_i is symmetric around 0, then you might want to resample from the $2n$ vector $X_i, -X_i, i = 1, \dots, n$.

Parameteric Bootstrap

- If you know F_0 is from a parametric family, say $\mathcal{E}(\lambda = \mu^{-1})$, then you may want to resample from $F(\lambda) = \mathcal{E}(\hat{\lambda})$ instead of the empirical distribution F_1 .
 - If you choose MLE, then it is $\hat{\lambda} = \frac{1}{\hat{\mu}} = \frac{1}{\bar{X}}$. So you resample from an exponential distribution with mean $\bar{X}$.
 - We usually focus on nonparametric bootstrap.
- The bootstrap principle again: The starting point is to find the definition of the functional $\theta(F_0)$.
 - It is often the solution $t = \theta(F_0)$ to $E[f(F_1, F_0; t) | F_0] = 0$.
 - Since we don't know F_0 , the bootstrap version is to estimate t by $\hat{t}$ s.t. $E[f(F_2, F_1; \hat{t}) | F_1] = 0$.
- Examples are bias reduction and confidence interval.

Bias Reduction

- Need $t = E(\theta(F_1) - \theta(F_0) | F_0)$. The bootstrap principle suggests estimating by $\hat{t} = E(\theta(F_2) - \theta(F_1) | F_1)$.
- For example,
 $\theta(F_0) = \mu^2 = (\int x dF_0(x))^2$, then $\theta(F_1) = \bar{X}^2 = (\int x dF_1(x))^2$.

$$E(\theta(F_1) | F_0) = E_{F_0} \left(\mu + n^{-1} \sum_{i=1}^n [X_i - \mu] \right)^2 = \mu^2 + n^{-1} \sigma^2$$

$$\implies t = n^{-1} \sigma^2 = O(n^{-1})$$

$$E(\theta(F_2) | F_1) = E_{F_1} \left(\bar{X} + n^{-1} \sum_{i=1}^n [X_i^* - \bar{X}] \right)^2 = \bar{X}^2 + n^{-1} \hat{\sigma}^2$$

$$\implies \hat{t} = n^{-1} \hat{\sigma}^2 \quad \text{where} \quad \hat{\sigma}^2 = n^{-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

- So the bootstrap bias-corrected estimate of μ^2 is:

$$\theta(F_1) - \hat{t} = 2\theta(F_1) - E(\theta(F_2)|F_1) = \bar{X}^2 - n^{-1}\hat{\sigma}^2$$

Its bias is:

$$E\left[\bar{X}^2 - n^{-1}\hat{\sigma}^2 - \mu^2|F_0\right] = n^{-1}\sigma^2 - n^{-1}\left(1 - n^{-1}\right)\sigma^2 = n^{-2}\sigma^2$$

So the bias is reduced by an order of $O(n^{-1})$, compared to the uncorrected estimate.

- For this problem, the one step bootstrap bias correction does not completely eliminate the bias.(It turns out bootstrap iteration will do)

Confidence Interval

- Look for a confidence interval of the form $(\hat{\theta} - t_1, \hat{\theta} + t_2)$ with approximate coverage probability of $1 - \alpha$:

$$\begin{aligned}P\left(\hat{\theta} - t_1 \leq \theta(F_0) \leq \hat{\theta} + t_2\right) &\rightarrow 1 - \alpha \\ \implies P\left(-t_2 \leq \hat{\theta} - \theta(F_0) \leq t_1\right) &\rightarrow 1 - \alpha.\end{aligned}$$

- The bootstrap version becomes

$$P_{F_1}\left(-\hat{t}_2 \leq \theta(F_2) - \theta(F_1) \leq \hat{t}_1\right) \rightarrow 1 - \alpha$$

- For equal-tail intervals, we split α so that

$$\begin{aligned}P_{F_1}\left(-\hat{t}_2 \leq \theta(F_2) - \theta(F_1)\right) &\rightarrow 1 - \alpha/2 \\ P_{F_1}\left(\theta(F_2) - \theta(F_1) \leq \hat{t}_1\right) &\rightarrow 1 - \alpha/2\end{aligned}$$

- In other words, $(-\hat{t}_2, \hat{t}_1)$ are the $(\alpha/2, 1 - \alpha/2)$ th quantiles of $\theta(F_2) - \theta(F_1)$ conditional on F_1 .

- Alternatively, $(-\hat{t}_1, \hat{t}_2)$ are the $(\alpha/2, 1 - \alpha/2)$ th quantiles of $\theta(F_1) - \theta(F_2)$ conditional on F_1 .
- The estimated confidence interval $(\hat{\theta} - \hat{t}_1, \hat{\theta} + \hat{t}_2)$, which is $(\theta(F_1) - \hat{t}_1, \theta(F_1) + \hat{t}_2)$, then becomes the $(\alpha/2, 1 - \alpha/2)$ th quantiles of $2\theta(F_1) - \theta(F_2)$ conditional on F_1 .
- This explains our earlier suggestion of using the $(\alpha/2, 1 - \alpha/2)$ th quantiles of the histogram of $2\hat{\theta} - \hat{\theta}_b, b = 1, \dots, B$ to form equal tail confidence intervals.
- You can also work out the recipe for symmetric confidence intervals.
- The regularity conditions for the validity of bootstrap confidence intervals (when properly constructed) are typically weaker than those for the validity of bootstrap asymptotic variance estimates.

- Usually the distribution function of $\theta(F_2) - \theta(F_1)$ conditional on F_1 is difficult to calculate, as difficult as $\theta(F_1) - \theta(F_0)$ conditional on $\theta(F_0)$.
- But at least the former can be simulated (since you know F_1), while the latter can't (since you don't know F_0).
- To simulate the distribution of $\theta(F_2) - \theta(F_1)$ conditional on F_1
 - (1) Independently draw B (a very big number, say 100,000) bootstrap resamples X_b^* , $b = 1, \dots, B$ from F_1 , where each $X_b^* = (X_{b1}^*, \dots, X_{bn}^*)^*$, each X_{bi}^* is independent draw from the empirical distribution.
 - (2) For each X_b^* , calculate $\theta_b^* = \theta(X_b^*)$. Then simply use the empirical distribution of X_b^* , or any smoothed version of it, to approximate the distribution of $\theta(F_2) - \theta(F_1)$ conditional on F_1 .

This approximation can be arbitrary close as $B \rightarrow \infty$.

Distribution of Test Statistics

- Similar to confidence interval construction.
- Consider a statistics (like OLS coefficient $\hat{\beta} - \beta_0$, t-statistics) $T_n = T_n(X_1, \dots, X_n)$, want to know its distribution function:

$$P_n(x, F_0) = P(T_n \leq x | X_1, \dots, X_n \sim iid F_0)$$

- But don't know F_0 , so use the bootstrap principle,

$$P_n(x, F_1) = P(T_n^* \leq x | X_1^*, \dots, X_n^* \sim iid F_1)$$

- Again when $P_n(x, F_1)$ can't be analytically computed, it can be approximated arbitrary well by for $T_{nb}^* = T_n(X_{b1}^*, \dots, X_{bn}^*)$,

$$P_n(x, F_1) \approx \frac{1}{B} \sum_{b=1}^B 1(T_{nb}^* \leq x)$$

- To construct T_n and T_n^* , be sure to follow the bootstrap principle properly, e.g. by replacing θ_0 under the null by $\hat{\theta}$ so that the distribution under the null is being approximated.

- Note again the schema in the bootstrap approximation.

$$P_n(x, F_0) \stackrel{1}{\approx} P_n(x, F_1) \stackrel{2}{\approx} \frac{1}{B} \sum_{b=1}^B 1(T_{nb}^* \leq x)$$

- 1 The statistical error: introduced by replacing F_0 with F_1 , the size of error as $n \rightarrow \infty$ can be analyzed through asymptotic theory, e.g. Edgeworth expansion.
 - 2 The numerical error: introduced by approximating F_1 using simulation. Should disappear as $B \rightarrow \infty$. It has nothing to do with n -asymptotics and statistical error.
- Similarly, standard error of T_n

$$\sigma^2(T_n) \approx \sigma^2(T_n^*) \approx \frac{1}{B} \sum_{b=1}^B \left(T_{nb}^* - \frac{1}{B} \sum_{b=1}^B T_{nb}^* \right)^2$$

- Similar to the parametric case, the nonparametric bootstrap sample (the histogram of $\hat{\theta}_b, b = 1, \dots, B$) can be used to (1) estimate asymptotic variance; (2) find new estimators; (3) form confidence intervals; (4) find distribution of test statistic. See above for (1), (2), (3).
- Suppose you want to test $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$, based on the test statistic $\hat{\theta} - \theta_0$. Reject when $|\hat{\theta} - \theta_0|$ larger than a critical value, which can be estimated by bootstrap.
- Find the $1 - \tau$ percentile of the histogram of $|\hat{\theta}_b - \hat{\theta}|, b = 1, \dots, B$:

$$\hat{q}_{1-\tau} = \inf\left\{q : \frac{1}{B} \sum_{b=1}^B 1\left(|\hat{\theta}_b - \hat{\theta}| \leq q\right) \geq 1 - \tau\right\}$$

- Reject if $|\hat{\theta} - \theta_0| \geq \hat{q}_{1-\tau}$ is an asymptotic size τ test.

- Important to use $|\hat{\theta}_b - \hat{\theta}|$ instead of $|\hat{\theta}_b - \theta_0|$ to form the bootstrap sample.
- Using $|\hat{\theta}_b - \theta_0|$ will remove power of the test.
- To achieve power, the goal is to estimate the distribution of the test statistic under H_0 even when the data is not generated under the null.
- Under the null, $\hat{\theta}$ is approximately θ_0 .
- Even if the data is generated under the alternative, the bootstrap distribution should still be approximated centered at zero.

The Pitfall of Bootstrap

- Whether the bootstrap works or not (in the consistency sense of whether $P(T_n^* \leq x|F_1) - P(T_n \leq x|F_0) \rightarrow 0$) need to be analyzed case by case.
- $\sqrt{n}$ consistent, asymptotically normal test statistics can be bootstrapped, but it is not known whether other things may work.
- Example of inconsistency, nonparametric bootstrap fails.

Take $F \sim U(0, \theta)$, and $X_{(1)}, \dots, X_{(n)}$ is the order statistics of the sample, so $X_{(n)}$ is the maximum. It is naturally to estimate θ using $X_{(n)}$.

- $\frac{\theta - X_{(n)}}{\theta}$ converges at rate n to $\mathcal{E}(1)$, since for $x > 0$:

$$\begin{aligned} P\left(n \frac{\theta - X_{(n)}}{\theta} > x\right) &= P\left(X_{(n)} < \theta - \frac{\theta x}{n}\right) = P\left(X_i < \theta - \frac{\theta x}{n}\right)^n \\ &= \left(\frac{1}{\theta} \left(\theta - \frac{\theta x}{n}\right)\right)^n = \left(1 - \frac{x}{n}\right)^n \xrightarrow{n \rightarrow \infty} e^{-x} \end{aligned}$$

In particular, the limiting distribution is continuous.

- But this is not the case for bootstrapped distribution, $X_{(n)}^*$.

The bootstrapped version is naturally $n(X_{(n)} - X_{(n)}^*)/X_{(n)}$. But

$$P\left(n \frac{X_{(n)} - X_{(n)}^*}{X_{(n)}} = 0\right) = \left(1 - \left(1 - \frac{1}{n}\right)^n\right) \xrightarrow{n \rightarrow \infty} (1 - e^{-1}) \approx 0.63$$

So there is a large probability mass at 0 in the limiting distribution of the bootstrap sample.

- It turns out that in this example parametric bootstrap would work although nonparametric bootstrap fails. But there are many examples where even parametric bootstrap will fail.
- An alternative to bootstrap, called subsampling, proposed by Romano(1998), is almost always consistent, as long as the subsample size m satisfies $m \rightarrow \infty$ and $m/n \rightarrow 0$. Serial correlation in time series also creates problem for naive nonparametric bootstrap. Subsampling is one solution.
- An alternative is to resample blocks instead of individual observations(Fitzenberg(1998) block bootstrap).
- However, both of these will only give consistency but not the 2nd order benefit of edgeworth expansion.

- So if in most cases bootstrap only works when asymptotic theory works, why use bootstrap?
- Some conceivable benefits are:
 - Don't want to waste time deriving asymptotic variance, although $\sqrt{n}$ consistency and asym normality is known. Let the computer do the job.
 - Avoid bandwidth selection in estimating var-cov of quantile regression type estimators. Bandwidth is needed for either kernel estimate of the conditional density $f(0|x_t)$ or for numerical derivatives.
 - For asymptotic pivotal statistics, bootstrapping is equivalent to automatically doing edgeworth expansion.

Exact Pivotal Statistics

- An exact (or asymptotic) pivotal statistics T_n is one whose (or asymptotic) distribution does not depend on unknown parameters $\forall n$.
- Denote pivotal statistics by T_n and nonpivotal ones by S_n .
- If we know that $F \sim N(\mu, \sigma^2)$, then
 - $S_n = \sqrt{n}(\bar{X} - \mu) \sim N(0, \sigma^2)$ is nonpivotal since σ^2 is unknown.
There is error in approximating the distribution of S_n by $N(0, \hat{\sigma}^2)$.
 - $T_n = \sqrt{n-1} \frac{(\bar{X} - \mu)}{\hat{\sigma}^2} \sim t_{n-1}$ for $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$.
No error in using the t_{n-1} table for the distribution of T_{n-1} .
- If T_n is exact pivotal, need not bootstrap at all. Either look up a t_{n-1} table or simulate. But most statistics are asymptotic pivotal.

Asymptotic Pivotal Statistics

- No matter what the underlying distribution F is, for t -statistics the CLT says $P(T_n \leq x) \xrightarrow{n \rightarrow \infty} \Phi(x)$, so it is asymptotically pivotal.
- But the CLT doesn't say how fast $P(T_n \leq x)$ tends to $\Phi(x)$.
- The Edgeworth expansion describes it:

$$P_n(x, F_0) \equiv P(T_n \leq x | F_0) = \Phi(x) + G(x, F_0) \frac{1}{\sqrt{n}} + O(n^{-1})$$

The bootstrap version is:

$$P_n(x, F_1) \equiv P(T_n^* \leq x | F_1) = \Phi(x) + G(x, F_1) \frac{1}{\sqrt{n}} + O_p(n^{-1})$$

- The Edgeworth expansion can be carried out up to many terms in power of $n^{-1/2}$. For example we can expand up to the 2nd term:

$$P_n(x, F_0) \equiv P(T_n \leq x | F_0) = \Phi(x) + G(x, F_0) \frac{1}{\sqrt{n}} + H(x, F_0) \frac{1}{n} + O(n^{-\frac{3}{2}})$$

Consider error in approximating $P_n(x, F_0)$:

- Error of CLT:

$$P_n(x, F_0) - \Phi(x) = G(x, F_0) \frac{1}{\sqrt{n}} + O(n^{-1}) = O\left(\frac{1}{\sqrt{n}}\right)$$

- Error of Bootstrap:

$$\begin{aligned} P_n(x, F_0) - P_n(x, F_1) &= G(x, F_0) \frac{1}{\sqrt{n}} - G(x, F_1) \frac{1}{\sqrt{n}} + O_p(n^{-1}) = \\ &= (G(x, F_0) - G(x, F_1)) \frac{1}{\sqrt{n}} + O_p(n^{-1}) = O_p(n^{-1}) \end{aligned}$$

since $\sqrt{n}(F_1 - F_0) = O_p(1)$, and assuming $G(x, F)$ is smooth and differentiable in the 2nd argument:

$$G(x, F_1) - G(x, F_0) = O_p(F_1 - F_0) = O_p\left(\frac{1}{\sqrt{n}}\right).$$

- If your sample size is 100, using CLT you commit an error of (roughly) 0.1, but using bootstrap only 0.01. Significant improvement?

- However, this improvement doesn't work for nonpitoval statistics such as S_n . By CLT

$$P(S_n \leq x) \xrightarrow{n \rightarrow \infty} \Phi\left(\frac{x}{\sigma}\right).$$

- The corresponding Edgeworth expansion is:

$$P_n(x, F_0) \equiv P(S_n \leq x | F_0) = \Phi\left(\frac{x}{\sigma}\right) + G(x/\sigma, F_0) \frac{1}{\sqrt{n}} + O(n^{-1})$$

The bootstrap version is:

$$P_n(x, F_1) \equiv P(S_n^* \leq x | F_1) = \Phi\left(\frac{x}{\hat{\sigma}}\right) + G(x/\hat{\sigma}, F_1) \frac{1}{\sqrt{n}} + O(n^{-1})$$

Consider error in approximating $P_n(x, F_0)$:

- Error of CLT: need to replace σ by $\hat{\sigma}$.

$$P_n(x, F_0) - \Phi(x/\hat{\sigma}) = \Phi(x/\sigma) - \Phi(x/\hat{\sigma}) + G(x/\sigma, F_0) \frac{1}{\sqrt{n}} + O(n^{-1}) = O\left(\frac{1}{\sqrt{n}}\right)$$

- Error of Bootstrap:

$$P_n(x, F_0) - P_n(x, F_1) = \Phi(x/\sigma) - \Phi(x/\hat{\sigma}) + G(x/\sigma, F_0) \frac{1}{\sqrt{n}} - G(x/\hat{\sigma}, F_1) \frac{1}{\sqrt{n}} + O_p(n^{-1}) = O_p(n^{-1/2})$$

This is because both $F_1 - F_0 = O_p\left(\frac{1}{\sqrt{n}}\right)$ and $\hat{\sigma} - \sigma = O_p\left(\frac{1}{\sqrt{n}}\right)$.

- No improvement compared to CLT. This is because now the 1st term $\Phi(x/\sigma)$ does not cancelled with $\Phi(x/\hat{\sigma})$.

- An additional implication of this is that bootstrapping provides better approximation to two sided symmetric test (or symmetric confidence interval) compared to one sided test (or confidence interval).
- Assume $G(x, F_0)$ is an even function in x .
- One-sided test: reject if $T_n \leq x$ (or $T_n > x$), the approximation error being:

$$\begin{aligned} P_n(x, F_0) - P_n(x, F_1) &= G(x, F_0) \frac{1}{\sqrt{n}} - G(x, F_1) \frac{1}{\sqrt{n}} + O_p(n^{-1}) \\ &= O_p(n^{-1}) \end{aligned}$$

- Two sided test: reject if $|T_n| \geq x \Leftrightarrow (T_n > x \cup T_n < -x)$, then

$$\begin{aligned}
 P(|T_n| > x) &= P(T_n > x) + P(T_n < -x) \\
 &= \left[1 - \Phi(x) - G(x, F_0) \frac{1}{\sqrt{n}} - H(x, F_0) \frac{1}{n} - O(n^{-3/2}) \right] \\
 &\quad + \left[\Phi(-x) + G(-x, F_0) \frac{1}{\sqrt{n}} + H(-x, F_0) \frac{1}{n} + O(n^{-3/2}) \right] \\
 &= 2\Phi(-x) - 2H(x, F_0) \frac{1}{n} + O(n^{-3/2})
 \end{aligned}$$

- So the approximation error is:

$$\begin{aligned}
 P(|T_n^*| > x | F_1) - P(|T_n| > x) &= 2[H(x, F_0) - H(x, F_1)] \frac{1}{n} + O(n^{-3/2}) \\
 &= O_p(n^{-3/2})
 \end{aligned}$$

Smaller by an order of $O_p(n^{-1/2})$.

Edgeworth Expansion

- We use the first term $G(x, F_0)$ to illustrate
- Let $X_1, \dots, X_n$ be iid $EX_i = 0, \text{Var}(X_i) = 1$. So $T_n = \sqrt{n}\bar{X}$
- Recall the characteristic function for T_n : by X_i iid assumption

$$\begin{aligned}\phi_{T_n}(t) &= Ee^{itT_n} = Ee^{it\frac{1}{\sqrt{n}}\sum_{i=1}^n X_i} = \left(Ee^{i\frac{t}{\sqrt{n}}X_i}\right)^n \\ &= \left[\phi_X\left(\frac{t}{\sqrt{n}}\right)\right]^n = e^{n \log \phi_X\left(\frac{t}{\sqrt{n}}\right)}\end{aligned}$$

- Taylor expand this around $\frac{t}{\sqrt{n}} = 0$:

$$\begin{aligned}
& n \log \phi_X \left(\frac{t}{\sqrt{n}} \right) \\
&= n \log \phi_X(0) + n \frac{\phi'_X(0)}{\phi_X(0)} \frac{t}{\sqrt{n}} + n \frac{1}{2} \left[\frac{\phi''_X(0)}{\phi_X(0)} - \frac{(\phi'_X(0))^2}{\phi_X(0)^2} \right] \left(\frac{t}{\sqrt{n}} \right)^2 \\
&+ n \frac{1}{3!} \left[\frac{\phi'''_X(0)}{\phi_X(0)} - 3 \frac{\phi'_X(0) \phi''_X(0)}{\phi_X(0)^2} + 2 \frac{\phi'_X(0)^3}{\phi_X(0)^3} \right] \left(\frac{t}{\sqrt{n}} \right)^3 + O \left(\frac{t}{\sqrt{n}} \right)^4
\end{aligned}$$

- Recall that $\phi_X(0) = 1$, $\phi'_X(0) = iEX = 0$, $\phi''_X(0) = i^2 EX^2 = -1$, $\phi'''_X(X) = i^3 EX^3 \equiv -i\mu^3$:

$$\begin{aligned}
n \log \phi_X \left(\frac{t}{\sqrt{n}} \right) &= -\frac{1}{2} t^2 - \frac{i}{6} \mu^3 \frac{t^3}{\sqrt{n}} + O \left(\frac{t^4}{n} \right) \\
\phi_{T_n}(t) &= e^{n \log \phi_X \left(\frac{t}{\sqrt{n}} \right)} = e^{-t^2/2} \exp \left(-\frac{i}{6} \mu^3 \frac{t^3}{\sqrt{n}} + O \left(\frac{t^4}{n} \right) \right) \\
&= e^{-t^2/2} \left[1 - \frac{i}{6} \mu^3 \frac{t^3}{\sqrt{n}} + O \left(n^{-1} \right) \right]
\end{aligned}$$

- Use the Inversion Formula: for $\phi_X(t) = Ee^{itX} = \int e^{itx} f(x) dx$, there is $f(x) = \frac{1}{2\pi} \int e^{-ixt} \phi_X(t) dt$
- For example, the characteristic function of $N(0, 1)$ is $e^{-t^2/2}$, so $e^{-t^2/2} = \int e^{itx} \phi(x) dx$. And also

$$\phi(x) = \frac{1}{2\pi} \int e^{-ixt} e^{-t^2/2} dt = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(t+ix)^2} dt = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}.$$

- Now applying this to $X = T_n$:

$$\begin{aligned} f_{T_n}(x) &= \frac{1}{2\pi} \int e^{-ixt} \phi_{T_n}(t) dt = \frac{1}{2\pi} \int e^{-ixt} e^{n \log \phi_X\left(\frac{t}{\sqrt{n}}\right)} dt \\ &= \frac{1}{2\pi} \int e^{-ixt} e^{-\frac{t^2}{2}} \left[1 - \frac{i}{6} \mu^3 \frac{t^3}{\sqrt{n}} + O(n^{-1}) \right] dt \\ &= \frac{1}{2\pi} \int e^{-ixt} e^{-\frac{t^2}{2}} dt - \frac{i}{6} \frac{\mu^3}{\sqrt{n}} \left(\frac{1}{2\pi} \int e^{-ixt} e^{-\frac{t^2}{2}} t^3 dt \right) \\ &= \frac{1}{2\pi} \int e^{-ixt} e^{-\frac{t^2}{2}} dt - \frac{i}{6} \frac{1}{(-i)^3} \frac{\mu^3}{\sqrt{n}} \left[\frac{d}{dx^3} \left(\frac{1}{2\pi} \int e^{-ixt} e^{-\frac{t^2}{2}} dt \right) \right] \\ &\quad + O(n^{-1}) = \phi(x) - \frac{1}{6} \frac{\mu^3}{\sqrt{n}} \phi'''(x) + O(n^{-1}) \end{aligned}$$

- So

$$P(T_n \leq x) = \int^x f_{T_n}(u) du = \Phi(x) - \frac{1}{6} \frac{\mu^3}{\sqrt{n}} \phi''(x) + O(n^{-1})$$

- So

$$G(x, F_0) = -\frac{\mu^3}{6} \phi''(x) = \frac{\mu^3}{6} (1 - x^2) \phi(x),$$

by noting that $\phi'(x) = -x\phi(x)$, and $\phi''(x) = -\phi(x) + x^2\phi(x)$.

- Note that $G(x, F_0)$ is an even function.

- Subsampling vs Bootstrap
- Based on Joe's book "Subsampling" (Dimitris N. Politis, Joseph P. Romano, Michael Wolf)
- Subsampling *works* whenever bootstrap *works*.
- Subsampling may work even when bootstrap does not work.
- When both works, bootstrap might be preferable to subsampling. The approximation error might be smaller.
- But it might or might not. Needs to be verified case by case.
- Michael Kosorok's "Bootstrapping empirical processes" is a more technical reference.

Bootstrap

- $R_n(x_n, \theta(P)) = \tau_n(\hat{\theta}_n - \theta(P))$
- Example:
 - $\hat{\theta}_n = \bar{X}_n, \tau_n = \sqrt{n}, \theta = EX = \mu(P)$
 - $\hat{\theta} = \min X_n, \tau_n = n, \theta(P) = \sup\{x : F(x) \leq 0\}$
- Define: $J_n(P)$, the distribution of $\tau_n(\hat{\theta}_n - \theta(P))$ under P . For real $\hat{\theta}_n$,

$$J_n(x, P) \equiv \text{Prob}_P\left(\tau_n(\hat{\theta}_n - \theta(P)) \leq x\right)$$

- Since P is unknown, $\theta(P)$ is unknown, and $J_n(x, P)$ is also unknown.

- The bootstrap estimate $J_n(x, P)$ by $J_n(x, \hat{P}_n)$, where $\hat{P}_n$ is a consistent estimate of P in some sense.
- For example, take $\hat{P}_n(x) = \frac{1}{n} \sum_{i=1}^n 1(X_i \leq x)$

$$\sup_x \left| \hat{P}_n(x) - P(x) \right| \xrightarrow{a.s.} 0$$

- Similarly estimate $(1 - \alpha)$ th quantile of $J_n(x, P)$ by $J_n(x, \hat{P}_n)$: i.e. Estimate $J_n^{-1}(x, P)$ by $J_n^{-1}(x, \hat{P}_n)$.
- Usually $J_n(x, \hat{P}_n)$ can't be explicitly calculated, use MC:

$$J_n(x, \hat{P}_n) \approx \frac{1}{B} \sum_{b=1}^B 1\left(\tau_n(\hat{\theta}_{n,b} - \hat{\theta}_n) \leq x\right)$$

for $\hat{\theta}_{n,b} = \hat{\theta}(X_{1,b}^*, \dots, X_{n,b}^*)$.

- When bootstrap works, for each x ,

$$J_n(x, \hat{P}_n) - J_n(x, P) \xrightarrow{P} 0 \implies J_n^{-1}(1 - \alpha, \hat{P}_n) - J_n^{-1}(1 - \alpha, P) \xrightarrow{P} 0$$

- When should Bootstrap “work”? Need local uniformity in weak convergence:

- Usually $J_n(x, P) \longrightarrow J(x, P)$.
- Usually $\hat{P}_n \xrightarrow{a.s.} P$ in some sense, say $\sup_x |\hat{P}_n(x) - P(x)| \xrightarrow{a.s.} 0$
- Suppose for each sequence P_n s.t. $P_n \rightarrow P$, say $\sup_x |P_n - P| \rightarrow 0$, it is also true that $J_n(x, P_n) \longrightarrow J(x, P)$, then it must be true that a.s. $J_n(x, \hat{P}_n) \longrightarrow J(x, P)$
- So it ends up having to show for $P_n \rightarrow P$, $J_n(x, P_n) \rightarrow J(x, P)$, use triangular array formulation.

Case When Bootstrap Works

- Sample mean with finite variance.
 - $\sup_x |\hat{F}_n(x) - F(x)| \xrightarrow{a.s.} 0.$
 - $\theta(\hat{F}_n) = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \theta(F) = E(X).$
 - $\sigma^2(\hat{F}_n) = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \xrightarrow{a.s.} \sigma^2(F) = \text{Var}(X).$
 - Use Linderberg-Feller for the triangular array, applied to the deterministic sequence of P_n such that:
 - 1) $\sup_x |P_n(x) - P(x)| \rightarrow 0;$
 - 2) $\theta(P_n) \rightarrow \theta(P);$
 - 3) $\sigma^2(P_n) \rightarrow \sigma^2(P),$it can be shown that $\sqrt{n}(\bar{X}_n - \theta(P_n)) \xrightarrow{d} N(0, \sigma^2)$ under P_n .
- Since $\hat{P}_n$ satisfies 1,2,3 a.s., therefore $J_n(x, \hat{P}_n) \xrightarrow{a.s.} J(x, P)$
- So “local uniformity” of weak convergence is satisfied here.

- Order Statistics: see earlier slides.
- Degenerate U-statistics:

Take $w(x, y) = xy$, $\theta(F) = \int \int w(x, y) dF(x) dF(y) = \mu(F)^2$.

$$\hat{\theta}_n = \theta(\hat{F}_n) = \frac{1}{n(n-1)} \sum_{i \neq j} X_i X_j$$

$$S(x) = \int xy dF(y) = x\mu(F)$$

- If $\mu(F) \neq 0$ it is known that

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N(0, 4\text{Var}(S(X))) = N\left(0, 4\left(\mu^2 EX^2 - \mu^4\right)\right)$$

The bootstrap works.

- But if $\mu(F) = 0 \implies \theta(F) = 0$:

$$\theta(\hat{F}_n) = \frac{1}{n(n-1)} \sum \sum_{i \neq j} x_i x_j = \bar{X}_n^2 - \frac{1}{n} \frac{1}{n-1} \sum_i (x_i - \bar{X}_n)^2 = \bar{X}_n^2 - \frac{S_n^2}{n}$$

$$n \left(\theta(\hat{F}_n) - \theta(F) \right) = n\bar{X}_n^2 - S_n^2 \xrightarrow{d} N(0, \sigma^2)^2 - \sigma^2$$

- However the bootstrap version of $n \left[\theta(\hat{F}_n^*) - \theta(\hat{F}_n) \right]$:

$$\begin{aligned} n \left(\left[\bar{X}_n^{*2} - \frac{1}{n} S_n^{*2} \right] - \left[\bar{X}_n^2 - \frac{1}{n} S_n^2 \right] \right) &= n\bar{X}_n^{*2} - S_n^{*2} - n\bar{X}_n^2 + S_n^2 \\ &\approx n \left(\bar{X}_n^{*2} - \bar{X}_n^2 \right) \\ &= \left[\sqrt{n} (\bar{X}_n^* - \bar{X}_n) \right]^2 + 2\sqrt{n} (\bar{X}_n^* - \bar{X}_n) \sqrt{n} \bar{X}_n \\ &\xrightarrow{d} N(0, \sigma^2)^2 + 2N(0, \sigma^2) \sqrt{n} \bar{X}_n \approx Z_1^2 + Z_1 \times Z_2 \end{aligned}$$

for Z_1 and Z_2 independent $N(0, \sigma^2)$.

Subsampling

- iid case: Y_i block of size b from $(X_1, \dots, X_n)$, $i = 1, \dots, q$, for $q = \binom{n}{b}$.
Let $\hat{\theta}_{n,b,i} = \hat{\theta}(Y_i)$ calculated with i th block of data.
- Use the empirical distribution of $\tau_b(\hat{\theta}_{n,b,i} - \hat{\theta})$ over the q pseudo-estimates to approximate the distribution of $\tau_n(\hat{\theta} - \theta)$:

$$\text{Approximate } J_n(x, P) = P\left(\tau_n(\hat{\theta}_n - \theta) \leq x\right)$$

$$\text{by } L_{n,b}(x) = q^{-1} \sum_{i=1}^q 1\left(\tau_b(\hat{\theta}_{n,b,i} - \hat{\theta}_n) \leq x\right)$$

- Claim: If $b \rightarrow \infty$, $b/n \rightarrow 0$, $\tau_b/\tau_n \rightarrow 0$, as long as $\tau_n(\hat{\theta} - \theta) \xrightarrow{d}$ something,

$$J_n(x, P) - L_{n,b}(x) \xrightarrow{P} 0$$

Different Motivation for Subsampling vs. Bootstrap Subsampling:

- Each subset of size b comes from the TRUE model. Since $\tau_n(\hat{\theta}_n - \theta) \xrightarrow{d} J(x, P)$, so as long as $b \rightarrow \infty$:

$$\tau_b(\hat{\theta}_b - \theta) \xrightarrow{d} J(x, P)$$

The distributions of $\tau_n(\hat{\theta}_n - \theta)$ and $\tau_b(\hat{\theta}_b - \theta)$ should be close.

- But $\tau_b(\hat{\theta}_b - \theta) = \tau_b(\hat{\theta}_b - \hat{\theta}_n) + \tau_b(\hat{\theta}_n - \theta)$. Since

$$\tau_b(\hat{\theta}_n - \theta) = O_p\left(\frac{\tau_b}{\tau_n}\right) = o_p(1)$$

The distributions of $\tau_b(\hat{\theta}_b - \theta)$ and $\tau_b(\hat{\theta}_b - \hat{\theta}_n)$ should be close.

- The distribution of $\tau_b(\hat{\theta}_b - \hat{\theta}_n)$ is estimated by the empirical distribution over $q = \binom{n}{b}$ pseudo-estimates.

Bootstrap:

- Recalculate the statistics from the ESTIMATED model $\hat{P}_n$.
- Given that $\hat{P}_n$ is close to P , hopefully $J_n(x, \hat{P}_n)$ is close to $J_n(x, P)$ (Or to $J(x, P)$, the limit distribution).
- But when

$$\hat{P}_n \rightarrow P \not\Rightarrow J_n(x, \hat{P}_n) \rightarrow J(x, P)$$

bootstrap fails.

Formal Proof of Consistency of Subsampling

- Assumptions: $\tau_n(\hat{\theta}_n - \theta) \xrightarrow{d} J(x, P)$, $b \rightarrow \infty$, $\frac{b}{n} \rightarrow 0$, $\frac{\tau_b}{\tau_n} \rightarrow 0$. Need to show: $L_{n,b}(x) - J(x, P) \xrightarrow{P} 0$.
- Since $\tau_b(\hat{\theta}_n - \theta) \xrightarrow{P} 0$, it is enough to show

$$U_{n,b}(x) = q^{-1} \sum_{i=1}^q 1\left(\tau_b(\hat{\theta}_{n,b,i} - \theta) \leq x\right) \xrightarrow{P} J(x, P).$$

$U_{n,b}(x)$ is a b th order U-statistics with kernel function bounded by $(-1, 1)$.

- $U_{n,b}(x) - J(x, P) = U_{n,b}(x) - EU_{n,b}(x) + EU_{n,b}(x) - J(x, P)$, it is enough to show

$$U_{n,b}(x) - EU_{n,b}(x) \xrightarrow{P} 0 \quad \text{and} \quad EU_{n,b}(x) - J(x, P) \rightarrow 0$$

- But

$$EU_{n,b}(x) - J(x, P) = J_b(x, P) \rightarrow 0$$

- Use Hoeffding exponential-type inequality (Serfling(1980), Thm A. p201):

$$\begin{aligned} P(U_{n,b}(x) - J_b(x, P) \geq \epsilon) &\leq \exp\left(-2\frac{n}{b}\epsilon^2 / [1 - (-1)]\right) \\ &= \exp\left(-\frac{n}{b}t^2\right) \rightarrow 0 \quad \text{as } \frac{n}{b} \rightarrow \infty. \end{aligned}$$

- So

$$\begin{aligned} L_{n,b}(x) - J(x, P) &= L_{n,b}(x) - U_{n,b}(x) + U_{n,b}(x) \\ &\quad - J_b(x, P) + J_b(x, P) - J(x, P) \xrightarrow{P} 0. \end{aligned}$$

Q.E.D.

Time Series

- Respect the ordering of the data to preserve correlation.

$$\hat{\theta}_{n,b,t} = \hat{\theta}_b(X_t, \dots, X_{t+b-1}), \quad q = T - b + 1.$$

$$L_{n,b}(x) = \frac{1}{q} \sum_{i=1}^q 1\left(\tau_b\left(\hat{\theta}_{n,b,t} - \hat{\theta}_n\right) \leq x\right)$$

- Assumption: $\tau_n(\hat{\theta}_n - \theta) \xrightarrow{d} J(x, P)$, $b \rightarrow \infty$, $\frac{b}{n} \rightarrow 0$, $\frac{\tau_b}{\tau_n} \rightarrow 0$, $\alpha(m) \rightarrow 0$.
- Result: $L_{n,b}(x) - J(x, P) \xrightarrow{p} 0$.
- Most difficult part: To show $\tau_n(\hat{\theta}_n - \theta) \xrightarrow{d} J(x, P)$.

- Can treat iid data as time series, or even using non-overlapping blocks $k = \lceil \frac{n}{b} \rceil$, but using $\binom{n}{b}$ more efficient.
- For example, if

$$\bar{U}_n(x) = k^{-1} \sum_{j=1}^k 1(\tau_b[R_{n,b,j} - \theta(P)] \leq x)$$

then

$$U_{n,b}(x) = E[\bar{U}_n(x) | \mathcal{X}_n] = E[1(\tau_b[R_{n,b,j} - \theta(P)] \leq x) | \mathcal{X}_n]$$

for $\mathcal{X}_n = (X_{(1)}, \dots, X_{(n)})$.

- $U_{n,b}(x)$ is better than $\bar{U}_n(x)$ since $\mathcal{X}_n$ is sufficient statistics for iid data.

- Hypothesis Testing: $T_n = \tau_n t_n(X_1, \dots, X_n)$,

$$G_n(x, P) = \text{Prob}_P(\tau_n \leq x) \xrightarrow{P \in P_0} J(x, P)$$

$$\hat{G}_{n,b}(x) = q^{-1} \sum_{i=1}^q 1(T_{n,b,i} \leq x) = q^{-1} \sum_{i=1}^q 1(\tau_b t_{n,b,i} \leq x)$$

As long as $b \rightarrow \infty$, $\frac{b}{n} \rightarrow 0$, $\tau_b/\tau_n \rightarrow 0$, then under $P \in P_0$:

$$\hat{G}_{n,b}(x) \rightarrow G(x, P)$$

If under $P \in P_1$, $T_n \rightarrow \infty$, then $\forall x$, $\hat{G}_{n,b}(x) \rightarrow 0$.

- Technically recentering is not necessary, but it will increase power.

Estimating the unknown rate of convergence

- Assume that $\tau_n = n^\beta$, for some unknown $\beta > 0$. Estimate β using different size of subsampling distribution.
- Key idea: Compare the shape of the empirical distributions of $\hat{\theta}_b - \hat{\theta}_n$ for different values of b to infer the value of β .
- Let $q = \binom{n}{b}$ for iid data, or $q = T - b + 1$ for time series data:

$$L_{n,b}(x|\tau_b) \equiv q^{-1} \sum_{a=1}^q 1 \left(\tau_b \left(\hat{\theta}_{n,b,a} - \hat{\theta}_n \right) \leq x \right)$$

$$L_{n,b}(x|1) \equiv q^{-1} \sum_{a=1}^q 1 \left(\hat{\theta}_{n,b,a} - \hat{\theta}_n \leq x \right)$$

- This implies

$$L_{n,b}(x|\tau_b) = L_{n,b} \left(\tau_b^{-1} x | 1 \right) \equiv t$$

- $x = L_{n,b}^{-1}(t|\tau_b) = \tau_b(\tau_b^{-1}x) = \tau_b L_{n,b}^{-1}(t|1)$
- Since $L_{n,b}(x|\tau_b) \xrightarrow{P} J(x, P)$, if $J(x, P)$ is continuous and increasing, it can be inferred that

$$L_{n,b}^{-1}(t|\tau_b) = J^{-1}(t, P) + o_p(1)$$

- Same as

$$\tau_b L_{n,b}^{-1}(t|1) = J^{-1}(t, P) + o_p(1)$$

- So

$$b^\beta L_{n,b}^{-1}(t|1) = J^{-1}(t, P) + o_p(1)$$

- Assuming $J^{-1}(t, P) > 0$, or $t > J(0, P)$, take log.

- For different b_1 and b_2 , then this becomes

$$\beta \log b_1 + \log \left(L_{n,b_1}^{-1}(t|1) \right) = \log J^{-1}(t, P) + o_p(1)$$

$$\beta \log b_2 + \log \left(L_{n,b_2}^{-1}(t|1) \right) = \log J^{-1}(t, P) + o_p(1)$$

- Different out the “fixed effect”

$$\beta (\log b_1 - \log b_2) = \log \left(L_{n,b_2}^{-1}(t|1) \right) - \log \left(L_{n,b_1}^{-1}(t|1) \right) + o_p(1)$$

- So estimate β by

$$\begin{aligned} \hat{\beta} &= (\log b_1 - \log b_2)^{-1} \left(\log \left(L_{n,b_2}^{-1}(t|1) \right) - \log \left(L_{n,b_1}^{-1}(t|1) \right) \right) \\ &= \beta + (\log b_1 - \log b_2)^{-1} \times o_p(1) \end{aligned}$$

- Take $b_1 = n^{\gamma_1}$, $b_2 = n^{\gamma_2}$, ($1 \geq \gamma_1 > \gamma_2 > 0$)

$$\hat{\beta} - \beta = ((\gamma_1 - \gamma_2) \log n)^{-1} o_p(1) = o_p((\log n)^{-1})$$

- How to know $t > J(0, P)$,

$$L_{n,b}(0|\tau_b) = L_{n,b}(0|1) = J(0, P) + o_p(1)$$

So estimating $J(0, P)$ not a problem.

- Alternatively, take $t_2 \in (0.5, 1)$, take $t_1 \in (0, 0.5)$

$$b^\beta \left(L_{n,b}^{-1}(t_2|1) - L_{n,b}^{-1}(t_1|1) \right) = J^{-1}(t_2|P) - J^{-1}(t_1|P) + o_p(1)$$

$$\beta \log b + \log \left(L_{n,b}^{-1}(t_2|1) - L_{n,b}^{-1}(t_1|1) \right) = \log \left(J^{-1}(t_2|P) - J^{-1}(t_1|P) \right) + o_p(1)$$

- $\hat{\beta} = (\log b_1 - \log b_2)^{-1}$
 $\left[\log \left(L_{n,b_2}^{-1}(t_2|1) - L_{n,b_2}^{-1}(t_1|1) \right) - \log \left(L_{n,b_1}^{-1}(t_2|1) - L_{n,b_1}^{-1}(t_1|1) \right) \right]$
- Take $b_1 = n^{\gamma_1}$, $b_2 = n^{\gamma_2}$ ($1 > \gamma_1 > \gamma_2 > 0$), $\hat{\beta} - \beta = o_p((\log n)^{-1})$.

Two Step Subsampling

- $\hat{\tau}_n = n^{\hat{\beta}}$

$$L_{n,b}(x|\hat{\tau}_b) = q^{-1} \sum_{a=1}^q 1\left(\hat{\tau}_b \left(\hat{\theta}_{n,b,a} - \hat{\theta}_n\right) \leq x\right)$$

Can show that

$$\sup_x \left| L_{n,b}(x|\hat{\tau}_b) - J(x, P) \right| \xrightarrow{P} 0.$$

- Problem: imprecise in small samples.
 - best choice of b gives $O(n^{-1/3})$ error rate.
 - Bootstrap pivotal statistics, when applicable, gives even better than $O(n^{-1/2})$ error rate.